

Divide Fractions

Lesson 2-3

Name:

Date:

Class:

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.

12 dots

$$12 \div 3 = 4$$

In $3/4 \div 1/2$, the dividend is $3/4$ — the total amount being split

Dividend

Write the definition:



$$12 \div 3 = 4$$

In $3/4 \div 1/2$, the divisor is $1/2$ — the size of each group

Divisor

Write the definition:

$$3/4 \rightarrow 4/3$$

flip top and bottom

The reciprocal of $2/3$ is $3/2$. To divide by $2/3$, multiply by $3/2$.

Reciprocal

Write the definition:

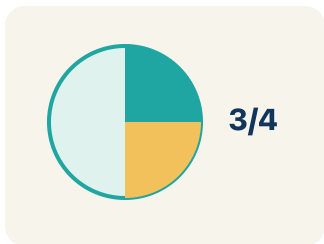


$$12 \div 3 = 4$$

In $3/4 \div 1/4 = 3$, the quotient is 3 — three $1/4$ -size pieces fit into $3/4$

Quotient

Write the definition:



$$8/12 \rightarrow \text{GCF is } 4 \rightarrow 8 \div 4 / 12 \div 4 = 2/3$$

Simplify

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can divide a fraction by a fraction by multiplying by the reciprocal and simplifying.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Dividend

Divisor

Reciprocal

Quotient

Simplify

1 In $\frac{3}{4} \div \frac{1}{2}$, the fraction $\frac{3}{4}$ being divided is the _____.

2 In $\frac{3}{4} \div \frac{1}{2}$, the fraction $\frac{1}{2}$ we divide by is the _____.

3 To divide by $\frac{1}{2}$, I multiply by its _____, which is 2.

4 The answer to a division problem is the _____.

5 Writing a fraction in its smallest equal form, like $\frac{4}{8} = \frac{1}{2}$, is to _____ it.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is the reciprocal of $\frac{3}{5}$?

- A. $\frac{5}{3}$
- B. $\frac{3}{5}$
- C. $\frac{5}{5}$
- D. $\frac{1}{3}$

Step 1 To find the reciprocal, flip the fraction.

Step 2 The reciprocal of $\frac{3}{5}$ is $\frac{5}{3}$.

 **Answer:** A. $\frac{5}{3}$

 We do — together

Solve this one with your class using the same steps.

Problem: What is $\frac{1}{2} \div \frac{1}{4}$?

- A. 2
- B. $\frac{1}{8}$
- C. 4
- D. $\frac{1}{2}$

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: What is $\frac{3}{4} \div \frac{1}{8}$?

- A. 6
- B. $\frac{3}{32}$
- C. $\frac{8}{3}$
- D. 24

Show your work:

| Try It

Solve on your own. Check the answer key when you are done.

1. Lock 2 — The vault dial reads a fraction problem: What is $(2/3) \div (1/6)$ in simplest form?

- A. 4
- B. $1/9$
- C. $2/18$
- D. $9/2$

Show your work:

2. Lock 3 — Decode the keypad: What is $(3/4) \div (1/2)$ in simplest form?

- A. $3/2$
- B. $3/8$
- C. $8/3$
- D. $2/3$

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

Is $\frac{3}{4} \div \frac{1}{2}$ the same as $\frac{1}{2} \div \frac{3}{4}$? Solve both and explain why or why not. What does this tell you about the order in division?

Sentence starter: $\frac{3}{4} \div \frac{1}{2} = \underline{\hspace{1cm}}$ and $\frac{1}{2} \div \frac{3}{4} = \underline{\hspace{1cm}}$. These are $\underline{\hspace{1cm}}$ because $\underline{\hspace{1cm}}$. This shows that division is $\underline{\hspace{1cm}}$.

Show your work:

Reflect — Exit Ticket

What is $\frac{4}{5} \div \frac{2}{5}$?

- A. 2
- B. $\frac{8}{25}$
- C. $\frac{2}{5}$
- D. $\frac{10}{5}$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Dividend
2. **Notes 2:** Divisor
3. **Notes 3:** Reciprocal
4. **Notes 4:** Quotient
5. **Notes 5:** Simplify
6. **We Try (notes):** A. $2 - 1/2 \div 1/4 = 1/2 \times 4/1 = 4/2 = 2$. Two $1/4$ -size pieces fit into $1/2$.
7. **You Try (notes):** A. $6 - 3/4 \div 1/8 = 3/4 \times 8/1 = 24/4 = 6$. Six $1/8$ -size pieces fit into $3/4$.
8. **Try It 1:** A. $4 - (2/3) \div (1/6) = (2/3) \times (6/1) = 12/3 = 4$. The dial unlocks at 4.
9. **Try It 2:** A. $3/2 - (3/4) \div (1/2) = (3/4) \times (2/1) = 6/4 = 3/2$. The keypad accepts $3/2$.
10. **Exit Ticket:** A. $2 - 4/5 \div 2/5 = 4/5 \times 5/2 = 20/10 = 2$. Two groups of $2/5$ fit into $4/5$.